

Flower Classification

<https://github.com/aiyshwary/flower-classification-project>

OVERVIEW

A CNN-based image classification project that identifies five flower types (Daisy, Dandelion, Rose, Sunflower, and Tulip) using a Kaggle Flowers Recognition dataset and a custom TensorFlow/Keras model pipeline.

IMPACT

Built an end-to-end notebook workflow that prepares the dataset, trains a custom CNN, and evaluates model performance on new flower species from images.

TECHNICAL WALKTHROUGH

Used the Kaggle Flowers Recognition dataset with five classes: Daisy, Dandelion, Rose, Sunflower, and Tulip. The dataset was downloaded, organized into class folders and loaded into the notebook for training and validation.

- i) Install dependencies: TensorFlow, OpenCV, scikit-learn, matplotlib, seaborn.
- ii) Place the dataset under `flowers/<class_name>/` to match the expected directory structure.

Images are resized to a consistent shape, normalized to $[0,1]$, and split into train/validation sets to ensure data diversity.

- i) Resize and normalize images to stabilize training.
- ii) Create stratified train/validation splits for balanced class coverage.

Implemented a custom CNN in Keras with stacked convolution + pooling blocks followed by dense layers and a softmax output.

- i) Convolutional blocks learn spatial features of petals and textures.
- ii) Dense layers map extracted features to 5-class softmax outputs.

Tracked training and validation curves to monitor convergence and prevent overfitting.

- i) Visualized accuracy/loss trends to validate model stability.
- ii) Evaluated class-wise performance using predictions on validation data.

The full workflow is implemented in the notebook for end-to-end reproducibility and easy experimentation.

KEY DETAILS

1. Organized the Kaggle Flowers Recognition dataset into class-specific folders for five flower categories.
2. Preprocessed images by resizing and normalizing pixel values, then created train/validation splits for model training.
3. Implemented a custom CNN architecture in TensorFlow/Keras for 5-class image classification.
4. Tracked training/validation accuracy and loss across epochs and visualized results with Matplotlib/Seaborn.
5. Packaged the workflow into a reproducible Jupyter Notebook for easy experimentation and iteration.
6. Uses OpenCV for image loading/processing and scikit-learn for train/test splitting alongside TensorFlow/Keras.
7. Serializes processed image data to a data.pickle file so subsequent runs skip re-reading raw images from disk.